Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin\theta$ and the length of the adjacent side.

Adjacent $= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin\theta = \frac{5}{13}$.

2 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos\theta$.

Adjacent $= \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$. $\cos\theta = \frac{8}{17}$.

3 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin\theta = \frac{24}{25}$, $\tan\theta = \frac{24}{7}$.

4 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin\theta$ and $\cos\theta$ for the angle θ opposite the side of length 20.

$\sin\theta = \frac{20}{29}$, $\cos\theta = \frac{21}{29}$.

5 **[Foundational]** • **Target: Special angles**

State the exact value of $\sin(30^\circ)$.

$\sin(30^\circ) = \frac{1}{2}$.

6 **[Foundational]** • **Target: Special angles**

State the exact value of $\cos(45^\circ)$.

$\cos(45^\circ) = \frac{\sqrt{2}}{2}$.

7 **[Foundational]** • **Target: Special angles**

State the exact value of $\tan(60^\circ)$.

$\tan(60^\circ) = \sqrt{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Special angles

State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

$\sin(90^\circ) = 1$ and $\cos(0^\circ) = 1$.

9 [Intermediate] • Target: Converting between radians and degrees

Convert 180° to radians.

$180^\circ = \pi$ radians.

10 [Intermediate] • Target: Converting between radians and degrees

Convert $\frac{5\pi}{6}$ radians to degrees.

$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$.

11 [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

12 [Intermediate] • Target: The Pythagorean identity

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin\theta = \frac{12}{13}$.

13 [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = 0.6$, find $\cos^2\theta$.

$\cos^2\theta = 1 - 0.36 = 0.64$.

14 [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\sin^2\theta + \cos^2\theta + 1$.

$\sin^2\theta + \cos^2\theta = 1$, so the expression equals $1 + 1 = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 2.01 Functions & Transformations

Simplify $\frac{1 - \cos^2\theta}{\sin\theta}$ (for $\sin\theta \neq 0$).

$1 - \cos^2\theta = \sin^2\theta$, so the expression equals $\frac{\sin^2\theta}{\sin\theta} = \sin\theta$.

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16 [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 3.01 Sequences & Series

Simplify $\tan\theta\cos\theta$ (for $\cos\theta \neq 0$).

$\tan\theta = \frac{\sin\theta}{\cos\theta}$, so $\tan\theta\cos\theta = \sin\theta$.

17 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

State the amplitude and period of $y = 5\sin(x)$.

Amplitude: 5. Period: 2π.

18 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 6.01 Limits & Derivatives

State the amplitude and period of $y = 3\cos(2x)$.

Amplitude: 3. Period: $\frac{2\pi}{2} = \pi$.

19 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 7.01 Integration & Applications

State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.

Amplitude: 2. Period: $\frac{2\pi}{1/3} = 6\pi$.

20 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 8.01 Exponentials & Logarithms

A sine graph has amplitude 4 and period π. Write its equation in the form $y = A\sin(Bx)$.

$A = 4$. $B = \frac{2\pi}{\pi} = 2$. So $y = 4\sin(2x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Right-triangle trigonometric ratios

In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin\theta$.

a. $\frac{17}{8}$

b. $\frac{8}{15}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$ ✓

$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}$.

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22 [Foundational] • Target: Right-triangle trigonometric ratios

In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin\theta$ and $\cos\theta$ (where θ is opposite the side of length 9).

a. $\sin\theta = \frac{9}{40}$, $\cos\theta = \frac{40}{9}$

b. $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$

c. $\sin\theta = \frac{41}{9}$, $\cos\theta = \frac{41}{40}$

d. $\sin\theta = \frac{9}{41}$, $\cos\theta = \frac{40}{41}$ ✓

Opposite $= 9$, adjacent $= 40$, hypotenuse $= 41$.

23 [Foundational] • Target: Graphs: amplitude and period

The graph shows $y = A\sin(x)$ for some A. Find A by reading the graph's amplitude.

a. $A = 1$

b. $A = 2\pi$

c. $A = 4$

d. $A = 2$ ✓

The curve's peak height (amplitude) is 2.

24 [Foundational] • Target: Graphs: phase shift and vertical shift

The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.

a. $y = 0$

b. $y = 2$ ✓

c. $y = 1$

d. $y = -2$

The curve oscillates evenly about $y = 2$.

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25 [Foundational] • Target: The unit circle: coordinates of key points

The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?

a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$ ✓

$\cos(60^\circ) = 0.5$, $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

26 [Foundational] • Target: Word problems: angles of elevation

A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.

a. 4

b. 12

c. 8

d. 11 ✓

$h = 12\sin(70^\circ) \approx 11$ m.

27 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos\theta$.

a. $\frac{9}{41}$

b. $\frac{9}{40}$

c. $\frac{41}{40}$

d. $\frac{40}{41}$ ✓

Adjacent $= \sqrt{41^2 - 9^2} = 40$; $\cos\theta = \frac{40}{41}$.

28 [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin\theta$.

a. $\frac{21}{20}$

b. $\frac{20}{29}$

c. $\frac{21}{29}$ ✓

d. $\frac{29}{21}$

Opposite $= \sqrt{29^2 - 20^2} = 21$; $\sin\theta = \frac{21}{29}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan\theta$.

a. $\frac{24}{7}$

b. $\frac{24}{25}$

c. $\frac{7}{25}$

d. $\frac{7}{24}$ ✓

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{7}{24}$.

30 **[Intermediate]** • **Target: Special angles**

State the exact values of $\cos(90^\circ)$ and $\sin(0^\circ)$.

a. $\cos(90^\circ) = 0, \sin(0^\circ) = 1$

b. $\cos(90^\circ) = 1, \sin(0^\circ) = 0$

c. Both equal 1

d. **Both equal** 0 ✓

$\cos(90^\circ) = 0$ and $\sin(0^\circ) = 0$.

31 **[Intermediate]** • **Target: The Pythagorean identity**

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\cos\theta$.

a. $\frac{3}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{5}{3}$

d. $\frac{9}{25}$

$\cos^2\theta = 1 - \frac{16}{25} = \frac{9}{25} \Rightarrow \cos\theta = \frac{3}{5}$.

32 **[Intermediate]** • **Target: Simplifying trigonometric expressions**

Simplify $\sin^2\theta + \cos^2\theta - 1$.

a. $2\sin^2\theta$

b. 0 ✓

c. 2

d. 1

By the Pythagorean identity, $\sin^2\theta + \cos^2\theta = 1$, so this is 0.

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33 [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\frac{1-\sin^2\theta}{\cos\theta}$ (for $\cos\theta \neq 0$).

a. $\frac{1}{\cos\theta}$

b. $\cos\theta$ ✓

c. $1-\cos\theta$

d. $\sin\theta$

$1-\sin^2\theta=\cos^2\theta$, so $\frac{\cos^2\theta}{\cos\theta}=\cos\theta$.

34 [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\frac{\sin\theta}{\tan\theta}$ (for $\tan\theta \neq 0$).

a. 1

b. $\sin\theta$

c. $\cos\theta$ ✓

d. $\frac{1}{\cos\theta}$

$\tan\theta=\frac{\sin\theta}{\cos\theta}$, so $\frac{\sin\theta}{\sin\theta/\cos\theta}=\cos\theta$.

35 [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $2\sin^2\theta+2\cos^2\theta$.

a. 1

b. 0

c. 2 ✓

d. $4\sin\theta\cos\theta$

$2(\sin^2\theta+\cos^2\theta)=2(1)=2$.

36 [Intermediate] • Target: Graphs: amplitude and period

A cosine graph has amplitude 3 and period 4π. Write its equation in the form $y=A\cos(Bx)$.

a. $y=3\cos(4\pi x)$

b. $y=4\pi\cos(3x)$

c. $y=3\cos\left(\frac{x}{2}\right)$ ✓

d. $y=3\cos(2x)$

$B=\frac{2\pi}{\text{period}}=\frac{2\pi}{4\pi}=\frac{1}{2}$.

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37 [Intermediate] • Target: Graphs: phase shift and vertical shift

The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the right and 3 units up. Write the new equation.

a. $y = \cos(x) + \frac{\pi}{4} + 3$

b. $y = \cos\left(x - \frac{\pi}{4}\right) + 3$ ✓

c. $y = \cos\left(x + \frac{\pi}{4}\right) + 3$

d. $y = \cos\left(x - \frac{\pi}{4}\right) - 3$

Right shift: replace x with $x - \frac{\pi}{4}$; up 3: add 3.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 6.01 Limits & Derivatives

State the vertical shift and midline of $y = 2\sin(x) - 4$.

a. Down 4, midline $y = 2$

b. Up 4, midline $y = 4$

c. **Down 4, midline** $y = -4$ ✓

d. Down 2, midline $y = -4$

Subtracting 4 shifts down 4; the midline is $y = -4$.

39 [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 7.01 Integration & Applications

The graph of $y = \sin(x)$ is shifted $\frac{\pi}{6}$ to the left. Write the new equation.

a. $y = \sin\left(\frac{\pi}{6} - x\right)$

b. $y = \sin(x) + \frac{\pi}{6}$

c. $y = \sin\left(x + \frac{\pi}{6}\right)$ ✓

d. $y = \sin\left(x - \frac{\pi}{6}\right)$

Left shift: replace x with $x + \frac{\pi}{6}$.

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40 [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 8.01 Exponentials & Logarithms

Solve $2\sin\theta = \sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 45^\circ$ only

b. $\theta = 45^\circ$ **or** $\theta = 135^\circ$ ✓

c. $\theta = 45^\circ$ or $\theta = 225^\circ$

d. $\theta = 60^\circ$ or $\theta = 120^\circ$

$\sin\theta = \frac{\sqrt{2}}{2}$: $\theta = 45^\circ$ or 135°.

41 [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 9.01 Statistics & Probability

Solve $\sqrt{3}\tan\theta = 3$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 60^\circ$ or $\theta = 300^\circ$

b. $\theta = 30^\circ$ or $\theta = 210^\circ$

c. $\theta = 60^\circ$ **or** $\theta = 240^\circ$ ✓

d. $\theta = 60^\circ$ only

$\tan\theta = \sqrt{3}$: $\theta = 60^\circ$ or 240° (tangent positive in Q1 and Q3).

42 [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 1.01 Algebra & Equations

Solve $2\cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 30^\circ$ or $\theta = 330^\circ$

b. $\theta = 60^\circ$ or $\theta = 240^\circ$

c. $\theta = 60^\circ$ only

d. $\theta = 60^\circ$ **or** $\theta = 300^\circ$ ✓

$\cos\theta = 0.5$: $\theta = 60^\circ$ or 300°.

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43 [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 2.01 Functions & Transformations

Solve $\tan\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 45^\circ$ **or** $\theta = 225^\circ$ ✓

b. $\theta = 45^\circ$ only

c. $\theta = 45^\circ$ or $\theta = 315^\circ$

d. $\theta = 45^\circ$ or $\theta = 135^\circ$

Tangent is positive in Q1 and Q3: $\theta = 45^\circ$ or 225°.

44 [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 3.01 Sequences & Series

Solve $2\sin\theta + 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 210^\circ$ or $\theta = 300^\circ$

b. $\theta = 150^\circ$ or $\theta = 330^\circ$

c. $\theta = 30^\circ$ or $\theta = 150^\circ$

d. $\theta = 210^\circ$ **or** $\theta = 330^\circ$ ✓

$\sin\theta = -0.5$: $\theta = 210^\circ$ or 330°.

45 [Challenging] • Target: Complementary angle relationships • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

a. 0.643

b. 0.766 ✓

c. 1.306

d. 0.234

$\sin(50^\circ) = \sin(90^\circ - 40^\circ) = \cos(40^\circ) \approx 0.766$.

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